



Yakeen Plus (2025)

Practice Test - 20

DURATION : 200 Minutes

DATE : 02/01/2025

M. MARKS : 720

Topics Covered

Physics :	Electrostatics, Current Electricity, Magnetic Effects of Current and Magnetism, Electromagnetic Induction and Alternating Currents, Electromagnetic Waves
Chemistry :	Thermodynamics and Thermochemistry, Chemical Kinetics, Alcohols, Ethers and Phenols, Aldehydes, Ketones and Carboxylic Acids, The p-Block Elements (Group 13,14,15, 16, 17, 18), The d and f-Block Elements
Biology :	(Botany) : Sexual Reproduction in Flowering Plant, Principle of Inheritance and Variation, Microbes in Human Welfare (Zoology) : Human Reproduction, Reproductive Health, Evolution

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Botany**) & Section IV (**Zoology**) and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

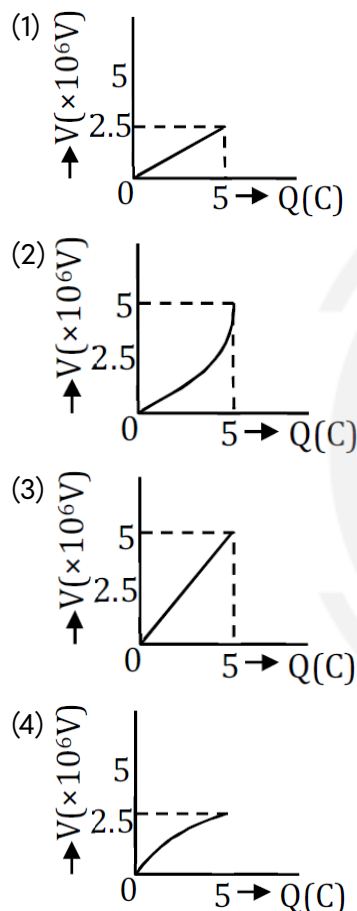
Practice Test-20

Yakeen Plus (2025)

Q1

SECTION - A (PHYSICS)

A condenser of $2 \mu F$ capacitance is charged steadily from 0 to 5 C. which of the following graph represents correctly the variation of potential difference (V) across it's plates with respect to the charge (Q) on the condenser?



- Q2** Two charged particles, having same kinetic energy, are allowed to pass through a uniform magnetic field perpendicular to the direction of motion. If the ratio of radii of their circular paths is 6 : 5 and their respective masses ratio is 9 : 4. Then, the ratio of their charges will be:
- (1) 8 : 5 (2) 5 : 4

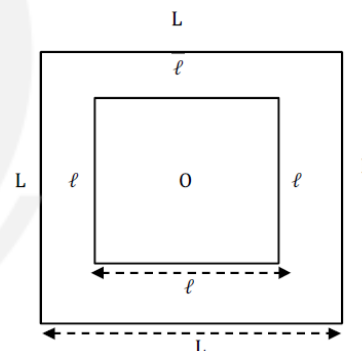
(3) 5 : 3

(4) 8 : 7

Q3 To increase the resonant frequency in series LCR circuit,

- (1) source frequency should be increased
 (2) another resistance should be added in series with the first resistance
 (3) another capacitor should be added in series with the first capacitor
 (4) the source frequency should be decreased

Q4 A small square loop of wire of side l is placed inside a large square loop of wire L ($L \gg l$). Both loops are coplanar and their centres coincide at point O as shown in figure. The mutual inductance of the system is:



- (1) $\frac{2\sqrt{2}\mu_0 L^2}{\pi l}$ (2) $\frac{\mu_0 L^2}{2\sqrt{2}\pi L}$
 (3) $\frac{2\sqrt{2}\mu_0 l^2}{\pi L}$ (4) $\frac{\mu_0 L^2}{2\sqrt{2}\pi l}$

Q5 The rms value of conduction current in a parallel plate capacitor is $6.9 \mu A$. The capacity of this capacitor, if it is connected to 230 V ac supply with an angular frequency of 600 rad/s will be:

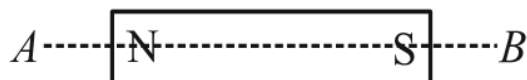
- (1) 5pF (2) 50 pF
 (3) 100 pF (4) 200 pF


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[iOS App](#)
[PW Website](#)

- Q6** The electric current in a circular coil of 2 turns produces a magnetic induction B_1 at its centre. The coil is unwound and is rewound into a circular coil of 5 turns and the same current produces a magnetic induction B_2 at its centre. The ratio of $\frac{B_2}{B_1}$ is:

- (1) $\frac{5}{2}$ (2) $\frac{25}{4}$
 (3) $\frac{5}{4}$ (4) $\frac{25}{2}$

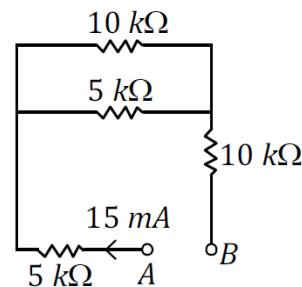
- Q7** If a bar magnet is cut identically into two parts along AB as shown in figure.



Choose the **incorrect** option.

- (1) Magnetic moment of each parts remains same as before cutting
 (2) Pole strength gets half after cutting
 (3) Magnetic moment gets half after cutting for each part
 (4) None of these
- Q8** Light wave traveling in air along x-direction is given by $E_y = 540 \sin \pi \times 10^4 (x - ct) Vm^{-1}$. Then, the peak value of magnetic field of wave will be (Given $c = 3 \times 10^8 ms^{-1}$)
- (1) $18 \times 10^{-7} T$
 (2) $54 \times 10^{-7} T$
 (3) $54 \times 10^{-8} T$
 (4) $18 \times 10^{-8} T$
- Q9** An electron with energy 0.1 keV moves at right angle to the earth's magnetic field of $1 \times 10^{-4} Wbm^{-2}$. The frequency of revolution of the electron will be (Takes mass of electron = $9.0 \times 10^{-31} kg$)
- (1) $1.6 \times 10^5 Hz$
 (2) $5.6 \times 10^5 Hz$
 (3) $2.8 \times 10^6 Hz$
 (4) $1.8 \times 10^6 Hz$

- Q10** A current of 15 mA flows in the circuit as shown in figure. The value of potential difference between the points A and B will be :



- (1) 50 V (2) 75 V
 (3) 150 V (4) 275 V

- Q11** The maximum error in the measurement of resistance, current and time for which current flows in an electrical circuit are 1%, 2% and 3% respectively. The maximum percentage error in the detection of the dissipated heat will be :

- (1) 2 (2) 4
 (3) 6 (4) 8

- Q12** In a series LR circuit $X_L = R$ and power factor of the circuit is P_1 , when capacitor with capacitance C such that $X_L = X_C$ in put in series, the power factor becomes P_2 The ratio $\frac{P_1}{P_2}$ is :

- (1) $\frac{1}{2}$ (2) $\frac{1}{\sqrt{2}}$
 (3) $\frac{\sqrt{3}}{\sqrt{2}}$ (4) 2 : 1

- Q13** A charge particle is moving in a uniform magnetic field $(2\hat{i} + 3\hat{j}) T$. If it has an acceleration of $(\alpha\hat{i} - 4\hat{j}) m/s^2$, then the value of α will be:

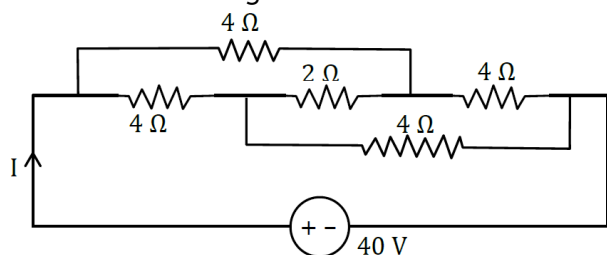
- (1) 3 (2) 6
 (3) 12 (4) 2

- Q14** B_X and B_Y are the magnetic fields at the center of two coils X and Y respectively, each carrying equal current. If coil X has 200 turns and 20 cm radius and coil Y has 400 turns and 20 cm radius, the ratio of B_X and B_Y is:

- (1) 1 : 1 (2) 1 : 2
 (3) 2 : 1 (4) 4 : 1



Q15 The current I in the given circuit will be:



- (1) 10 A (2) 20 A
(3) 4 A (4) 40 A

Q16 The total charge on the system of capacitors $C_1 = 1\mu\text{F}$, $C_2 = 2\mu\text{F}$, $C_3 = 4\mu\text{F}$ and $C_4 = 3\mu\text{F}$ Connected in parallel is: (Assume a battery of 20 V is connected to the combination)

- (1) 200 μC (2) 200 C
(3) 10 μC (4) 10 C

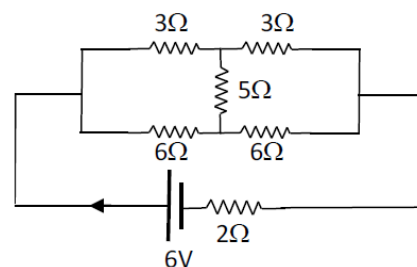
Q17 Two uniformly charged spherical conductors A and B of radii 5 mm and 10 mm are separated by a distance of 2 cm. If the spheres are connected by a conducting wire, then in equilibrium condition, the ratio of the magnitudes of the electric fields at the surface of the sphere A and B will be:

- (1) 1 : 2 (2) 2 : 1
(3) 1 : 1 (4) 1 : 4

Q18 The oscillating magnetic field in a plane electromagnetic wave is given by $B_y = 5 \times 10^{-6} \sin 1000\pi (5x - 4 \times 10^8 t) T$. The amplitude of electric field will be:

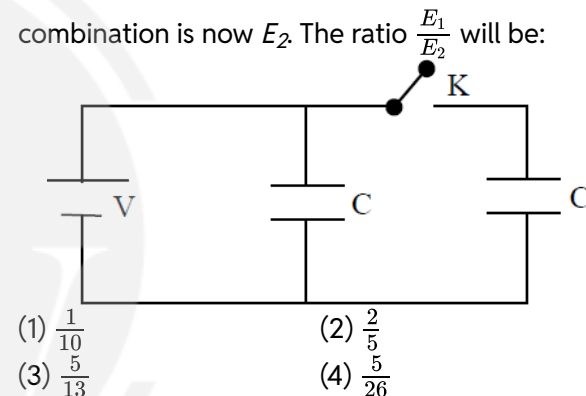
- (1) $15 \times 10^2 \text{Vm}^{-1}$
(2) $5 \times 10^{-6} \text{Vm}^{-1}$
(3) $16 \times 10^{12} \text{Vm}^{-1}$
(4) $4 \times 10^2 \text{Vm}^{-1}$

Q19 A battery of 6 V is connected to the circuit as shown below. The current I drawn from the battery is:



- (1) 1 A (2) 2 A
(3) $\frac{6}{11}$ A (4) $\frac{4}{3}$ A

Q20 A source of potential difference V is connected to the combination of two identical capacitors as shown in the figure. When key 'K' is closed, the total energy stored across the combination is E_1 . Now key 'K' is opened and dielectric of dielectric constant 5 is introduced between the plates of the capacitors. The total energy stored across the combination is now E_2 . The ratio $\frac{E_1}{E_2}$ will be:



- (1) $\frac{1}{10}$ (2) $\frac{2}{5}$
(3) $\frac{5}{13}$ (4) $\frac{5}{26}$

Q21 The radius of a circular current carrying coil is R . At what distance from the centre of the coil on its axis, the intensity of magnetic field will be $\frac{1}{2\sqrt{2}}$ times that at the centre?

- (1) $2R$ (2) $\frac{3R}{2}$
(3) R (4) $\frac{R}{2}$

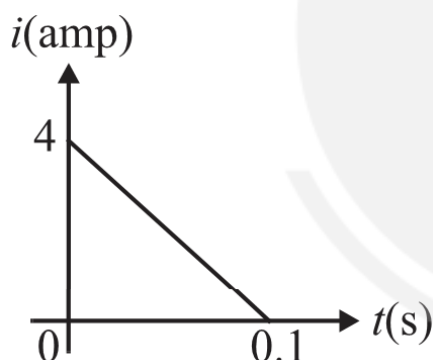
Q22 A coil of resistance 400Ω is placed in a magnetic field. If the magnetic flux ϕ (Wb) linked with the coil varies with time t (sec) as $\phi = 50t^2 + 4$. The current in the coil at $t = 2$ sec is:

- (1) 0.5 A (2) 0.1 A
(3) 2 A (4) 1 A



- Q23** Consider the following statements.
 (A) The drift velocity of electrons decreases with the increase in the temperature of conductor.
 (B) the drift velocity is inversely proportional to the area of cross-section of given conductor.
 (C) The drift velocity does not depend on the applied potential difference to the conductor.
 (D) The drift velocity of electron is inversely proportional to the length of the conductor.
 (E) The drift velocity of electrons increases with the increase in temperature of conductor.
 Choose the **correct** answer from the option given below:
 (1) (A) and (B) only
 (2) (A) and (D) only
 (3) (B) and (E) only
 (4) (B) and (C) only

- Q24** In a coil of resistance $10\ \Omega$, the induced current developed by changing magnetic flux through it, is shown in figure find a magnitude of change in flux through the coil in weber:



- (1) 8
(3) 6
(2) 2
(4) 4
- Q25** A current of i is flowing in an equilateral triangle of side a . The magnetic induction at the centroid will be:
 (1) $\frac{\mu_0 i}{3\sqrt{3}\pi a}$
 (2) $\frac{3\mu_0 i}{2\pi a}$
 (3) $\frac{5\sqrt{2}\mu_0 i}{3\pi a}$
 (4) $\frac{9\mu_0 i}{2\pi a}$
- Q26** Two capacitors, each having capacitance 40 mF are connected in series. The space between one

of the capacitors is filled with dielectric material of dielectric constant K such that the equivalence capacitance of the system became 24 mF. The value of K will be:

- (1) 1.5
(2) 2.5
(3) 1.2
(4) 3

- Q27** A wire of resistance R_1 is drawn out so that its length is increased by twice of its original length. The ratio of new resistance to original resistance is:

- (1) 9 : 1
(2) 1 : 9
(3) 4 : 1
(4) 3 : 1

- Q28** The current sensitivity of a galvanometer can be increases by:

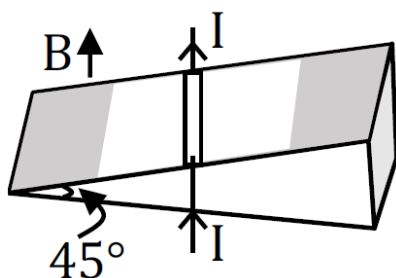
- (A) Decreasing the number of turns
 (B) Increasing the magnetic field
 (C) Decreasing the area of the coil
 (D) Decreasing the torsional constant of the spring

Choose the most appropriate answer from the options given below:

- (1) (B) and (C) only
 (2) (C) and (D) only
 (3) (A) and (C) only
 (4) (B) and (D) only

- Q29** As shown in the figure, a metallic rod of linear density 0.45 kg m^{-1} is lying horizontally on a smooth inclined plane which makes an angle of 45° with the horizontal. The minimum current flowing in the rod required to keep it stationary, when 0.15 T magnetic field is acting on it in the vertical upward direction, will be: {Use $g = 10\text{ m/s}^2$ }





- (1) 30 A (2) 15 A
(3) 10 A (4) 3 A

Q30 The equation of current in a purely inductive circuit is $5\sin(49\pi t - 30^\circ)$ A. If the inductance is 30 mH then the equation for the voltage across the inductor (in V), will be: {Let $\pi = 22/7$ }

- (1) $1.47 \sin(49\pi t - 30^\circ)$
(2) $1.47 \sin(49\pi t + 60^\circ)$
(3) $23.1 \sin(49\pi t - 30^\circ)$
(4) $23.1 \sin(49\pi t + 60^\circ)$

Q31 A slab of dielectric constant K has the same cross-sectional area as the plates of a parallel plate capacitor and thickness $\frac{3}{4}d$, where d is the separation of the plates. The capacitance of the capacitor when the slab is inserted between the plates will be: (Given C_0 = capacitance of capacitor with air as medium between plates.)

- (1) $\frac{4KC_0}{3+K}$ (2) $\frac{3KC_0}{3+K}$
(3) $\frac{3+K}{4KC_0}$ (4) $\frac{K}{4+K}$

Q32 Given below are two statements:

Statement I: A charge is moving in a uniform magnetic field. The charge may experience no magnetic force.

Statement II: Kinetic energy of a charged particle moving in a uniform magnetic field may change.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q33 For a diamagnetic substance (symbols have their usual meanings)

- (1) $\chi_m > 0, \mu_r > 1$
(2) $\chi_m < 0, \mu_r > 1$
(3) $\chi_m < 0, \mu_r < 1$
(4) $\chi_m > 0, \mu_r < 1$

Q34 A charged particle with charge q is moving in a uniform magnetic field. If this particle makes any angle (other than 0° , 90° and 180°) with the magnetic field then its path will be:

- (1) Circular (2) Straight line
(3) Helical (4) Parabolic

Q35 The magnetic field at the center of current carrying circular loop is B_1 . The magnetic field at a distance of $\sqrt{3}$ times radius of the given circular loop from the center on its axis is B_2 . The value of $\frac{B_1}{B_2}$ will be:

- (1) 9 : 4 (2) $12 : \sqrt{15}$
(3) 8 : 1 (4) $5 : \sqrt{3}$

Q36

SECTION - B (PHYSICS)

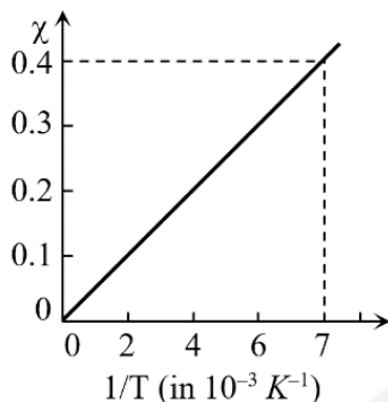
Two similar coils of radius R are lying concentrically with their planes at right angles to each other. The currents flowing in them are I and $2I$, respectively. The resultant magnetic field at the centre will be:

- (1) $\frac{\mu_0 I}{2R}$
(2) $\frac{\mu_0 I}{R}$



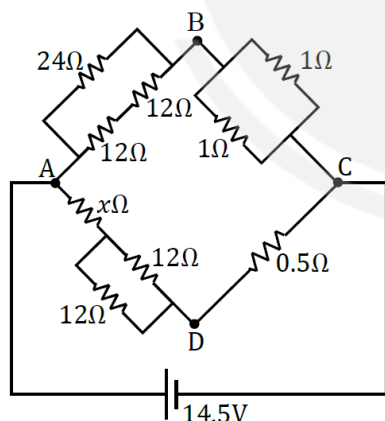
- (3) $\frac{\sqrt{5}\mu_0 I}{2R}$
 (4) $\frac{3\mu_0 I}{2R}$

Q37 The $\chi - (1/T)$ graph for an alloy of paramagnetic nature is shown in figure. The curie constant is, then:



- (1) 57 K
 (2) 2.8×10^{-3} K
 (3) 570 K
 (4) 17.5×10^{-3} K

Q38 The value of unknown resistance (x) for which the potential difference between B and D will be zero in the arrangement shown, is:



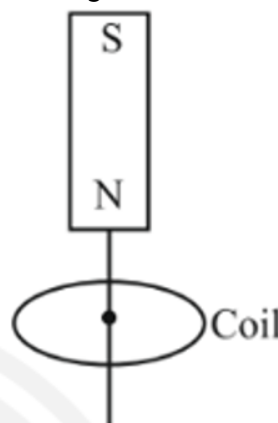
- (1) 6Ω
 (2) 9Ω
 (3) 42Ω
 (4) 3Ω

Q39 The number of electrons flowing per second in the filament of a 110 W bulb operating at 220 V is: (Given $e = 1.6 \times 10^{-19}\text{C}$)

- (1) 1.25×10^{19}

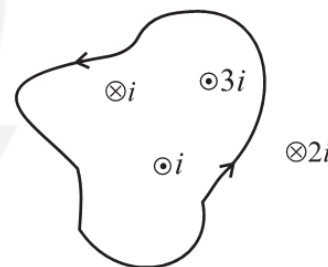
- (2) 6.25×10^{17}
 (3) 6.25×10^{18}
 (4) 31.25×10^{17}

Q40 A bar magnet is released along the vertical axis of the conducting coil. The acceleration of the bar magnet is:



- (1) Greater than g
 (2) Less than g
 (3) Equal to g
 (4) Zero

Q41 The value of $\oint \vec{B} \cdot d\vec{l}$ for the loop as shown in the figure will be:



- (1) $2\mu_0 i$
 (2) $3\mu_0 i$
 (3) $\mu_0 i$
 (4) $5\mu_0 i$

Q42 Two identical conducting spheres P and S with charge Q on each, repel each other with a force 16 N. A third identical uncharged conducting sphere R is successively brought in contact with the two spheres. The new force of repulsion between P and S at same separation is:

- (1) 4 N
 (2) 1 N
 (3) 6 N
 (4) 12 N



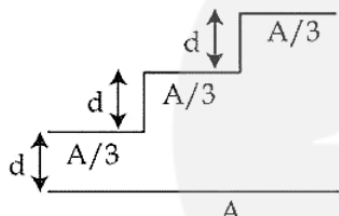
Q43 A coil having an area A_0 is placed in a magnetic field which changes from B_0 to $4B_0$ in a time interval t . The emf induced in the coil will be:

- (1) $\frac{3A_0B_0}{A_0t}$ (2) $\frac{4A_0B_0}{t}$
 (3) $\frac{3B_0}{A_0t}$ (4) $\frac{4B_0}{A_0t}$

Q44 A bulb and a capacitor are connected in series across an AC supply. A dielectric is then placed between the plates of the capacitor. The glow of the bulb:

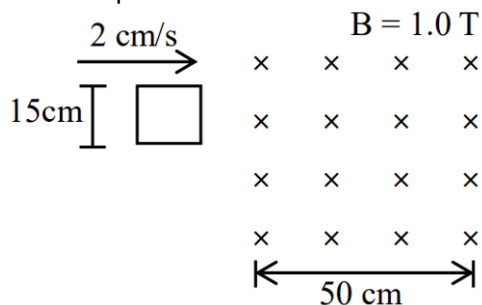
- (1) remains same (2) decrease
 (3) increase (4) becomes zero

Q45 A capacitor is made of a flat plate of area A and a second plate having a stair-like structure as shown in figure. If the area of each stair is $\frac{A}{3}$ and the height is d , the capacitance of the arrangement is:



- (1) $\frac{18\epsilon_0 A}{11d}$ (2) $\frac{11\epsilon_0 A}{20d}$
 (3) $\frac{13\epsilon_0 A}{17d}$ (4) $\frac{11\epsilon_0 A}{18d}$

Q46 A square loop of side 15 cm being moved towards right at a constant speed of 2 cm/s as shown in figure. The front edge enters the 50 cm wide magnetic field at $t = 0$. The value of induced emf in the loop at $t = 10$ s will be:



- (1) 4.5 mV (2) 3 mV
 (3) 0.3 mV (4) zero

Q47 An electric bulb rated 50 W – 200 V is connected across a 100 V supply. The power dissipation of the bulb is:

- (1) 12.5 W (2) 25 W
 (3) 100 W (4) 50 W

Q48 Match List-I with List-II

List-I	List-II
A Purely capacitive circuit	I.
B Purely inductive circuit	II.
C LCR series at resonance	III.
D LCR series circuit	IV.

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-II, D-III
 (2) A-I, B-IV, C-III, D-II
 (3) A-I, B-IV, C-II, D-III
 (4) A-IV, B-I, C-III, D-II

Q49 The magnetic moment of a bar magnet is 0.5 Am^2 . It is suspended in a uniform magnetic field of $8 \times 10^{-2} \text{ T}$. The work done in rotating it from its most stable to most unstable position is:

- (1) $16 \times 10^{-2} \text{ J}$
 (2) zero
 (3) $8 \times 10^{-2} \text{ J}$
 (4) $4 \times 10^{-2} \text{ J}$



Q50 In a plane electromagnetic wave, the directions of electric field and magnetic field are represented by $\hat{k}$ and $2\hat{i} - 2\hat{j}$ respectively. What is the unit vector along direction of propagation of the wave.

- (1) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ (2) $\frac{1}{\sqrt{5}}(2\hat{i} + \hat{j})$
 (3) $\frac{1}{\sqrt{5}}(\hat{i} + 2\hat{j})$ (4) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{k})$

Q51 **SECTION - A (CHEMISTRY)**

$K_2Cr_2O_7$ is:

- (1) orange-red coloured crystalline compound
 (2) moderately soluble in cold water
 (3) a powerful oxidising agent in acidic medium
 (4) all of these

Q52 Benzoyl chloride on reduction with $H_2/Pd-BaSO_4$ produces:

- (1) benzoic acid
 (2) benzyl alcohol
 (3) benzoyl sulphate
 (4) benzaldehyde

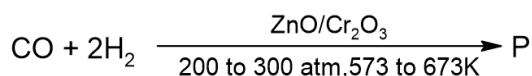
Q53 Chromium has the most stable oxidation state in acidic solution:

- (1) +2 (2) +5
 (3) +4 (4) +3

Q54 Which of the following halogen is liquid at room temperature?

- (1) Fluorine (2) Chlorine
 (3) Bromine (4) Iodine

Q55 Identify the product in the following reaction:



the product P is:

- (1) CH_3COOH (2) $HCHO$
 (3) CH_3OH (4) CH_3CH_2OH

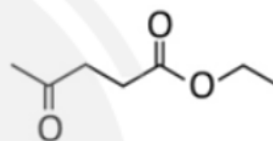
Q56 Activation energy of a forward reaction is found to be 58 kJ/mol while after adding a catalyst it lowers to 43 kJ/mol. What effect does it have on the activation energy of the backward reaction?

- (1) Increases by 30 kJ mol⁻¹
 (2) Decreases by 30 kJ mol⁻¹
 (3) Decrease by 15 kJ mol⁻¹
 (4) Increase by 15 kJ mol⁻¹

Q57 Reaction of Phenol with dilute nitric acid at low temperature gives:

- (1) o-nitrophenol and p-nitrophenol
 (2) m-nitrophenol
 (3) 2,4,6-trinitrophenol
 (4) both (1) and (2)

Q58 IUPAC name of the following compound is:



- (1) Ethyl 4-oxopentanoate
 (2) 3-oxoethylpentanoate
 (3) Ethyl 3-oxopentanoate
 (4) Ethyl 4-ketopentylester

Q59 Which of the following compounds will have the smallest pK_a value?

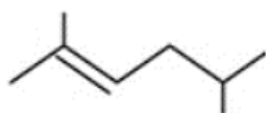
- (1) Benzoic acid
 (2) Formic acid
 (3) Acetic acid
 (4) Phenylacetic acid

Q60 Which of the lanthanides has +2 as most stable oxidation state?

- (1) Ce (2) Eu
 (3) Tb (4) Dy

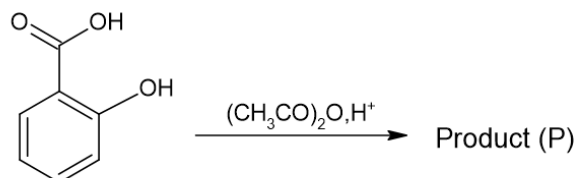
Q61 Reductive ozonolysis of the following compound gives:





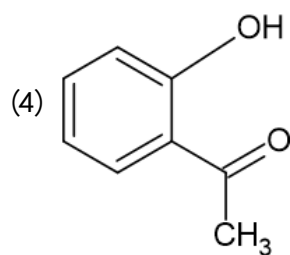
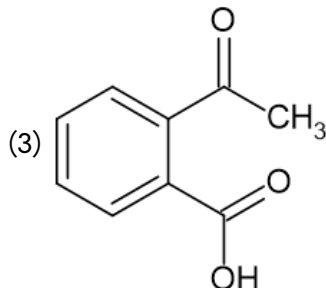
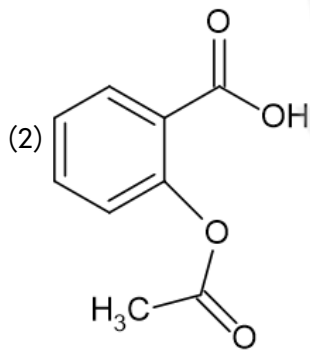
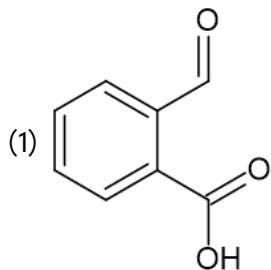
- (1) propan-2-one, 2-methylbutanal
 (2) butan-2-one, 3-methylhexanal
 (3) propan-2-one, 3-methylbutanal
 (4) butan-2-one, 2-methylbutanal

Q62 In the reaction,



Salicylic acid

Product (P) is:



Q63 Which of the following represent the **correct** structure a hemiacetal?

- (1)
- (2)
- (3)
- (4)

Q64 Benzaldehyde reacts with methyl amine to give:

- (1) $C_6H_5NH_2$ (2) $C_6H_5CH_2NH_2$
 (3) $C_6H_5CH=NCH_3$ (4) $C_6H_5CONH_2$

Q65 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Picric acid is a strong acid inspite of absence of carboxylic group.

Reason R: Three $-NO_2$ groups in picric acid activate the phenolate ion.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.



(4) Both A and R are true but R is NOT the correct explanation of A.

Q66 What will be the charge on Fe^{x+} if the magnetic moment is $\sqrt{24}$ B.M.?

- (1) +2 (2) +3
(3) Zero (4) +1

Q67 Phenylmagnesium bromide reacts with methanol to give:

- (1) a mixture of anisole and $\text{Mg}(\text{OH})\text{Br}$
(2) a mixture of benzene and $\text{Mg}(\text{OMe})\text{Br}$
(3) a mixture of toluene and $\text{Mg}(\text{OH})\text{Br}$
(4) a mixture of phenol and $\text{Mg}(\text{OMe})\text{Br}$

Q68 Name the gas that can readily decolourise acidified KMnO_4 solution.

- (1) SO_2 (2) N_2O_5
(3) P_2O_5 (4) CO_2

Q69 The reason for the stability of Gd^{3+} ion is:

- (1) half filled 4f-subshell
(2) completely filled 4f-subshell
(3) possesses the general electronic configuration of noble gases
(4) empty 4f-subshell

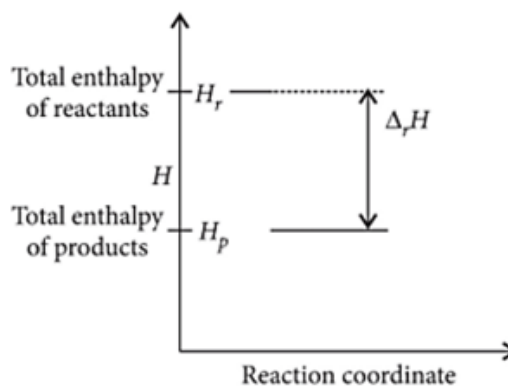
Q70 Phenol and ethanol can be distinguished by:

- (1) aqueous NaHCO_3 (2) Iodoform test
(3) sodium metal (4) aqueous Na_2CO_3

Q71 Which one of the following ions is the most stable in aqueous solution?

- (1) Mn^{2+} (2) Cr^{3+}
(3) V^{3+} (4) Ti^{3+}

Q72 In below diagram $\Delta_r H$ represents:

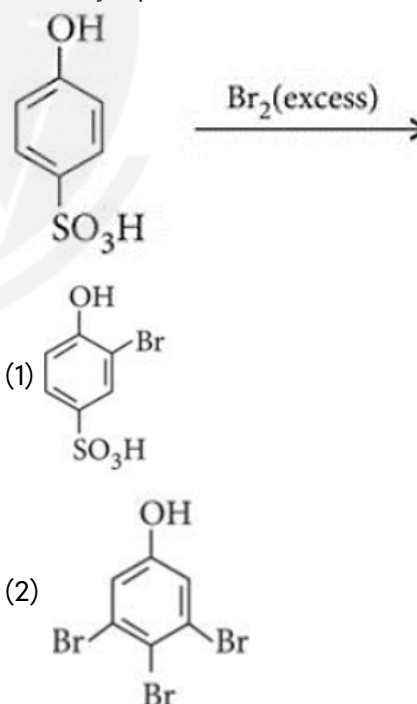


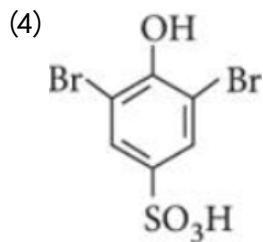
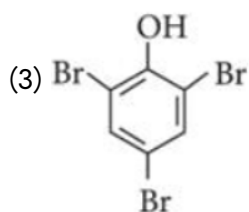
- (1) enthalpy for exothermic reaction.
(2) enthalpy for endothermic reaction.
(3) entropy for exothermic reaction.
(4) entropy for endothermic reaction.

Q73 The rate constant for a first order reaction is $4.606 \times 10^{-3} \text{ s}^{-1}$. The time required to reduce 2.0 g of the reactant to 0.2 g is:

- (1) 100 s (2) 200 s
(3) 500 s (4) 1000 s

Q74 The major product of the following reaction is:

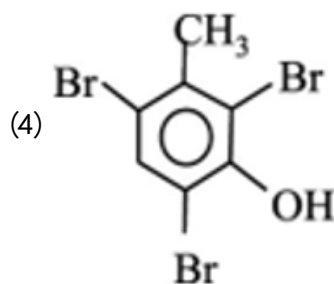
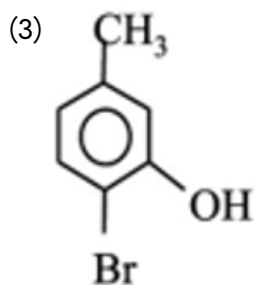
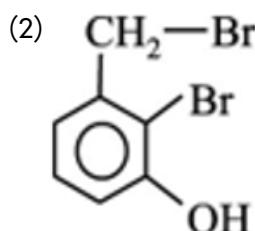
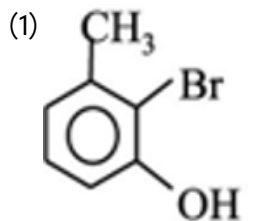




Q75 Hybridised state of bromine in bromine pentafluoride is:

- (1) sp^3d (2) dsp^2
(3) dsp^3 (4) sp^3d^2

Q76 m-Cresol on bromination gives:



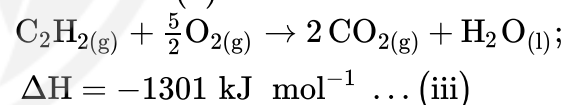
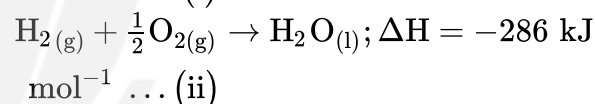
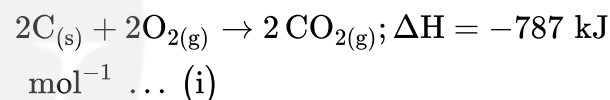
Q77 The elements having the maximum and minimum melting points among the members of group 13 respectively are:

- (1) B, Tl (2) Tl, B
(3) B, Ga (4) Al, Tl

Q78 Each of the following is **true** for white and red phosphorus **except** that they:

- (1) are both soluble in CS_2
(2) can be oxidised by heating in air
(3) consist of the same kind of atoms
(4) can be converted into one another

Q79 Given that,



Heat of formation of acetylene is:

- (1) $-1802 \text{ kJ mol}^{-1}$
(2) $+1802 \text{ kJ mol}^{-1}$
(3) -800 kJ mol^{-1}
(4) $+228 \text{ kJ mol}^{-1}$

Q80 For the reaction, $4KClO_3 \rightarrow 3KClO_4 + KCl$

If $\frac{-d[KClO_3]}{dt} = k_1$, $\frac{d[KClO_4]}{dt} = k_2$, $\frac{d[KCl]}{dt} = k_3$

the **correct** relation between k_1 , k_2 and k_3 is:

- (1) $k_1 = k_2 = k_3$
(2) $4k_1 = 3k_2 = k_3$
(3) $3k_1 = 4k_2 = 12k_3$
(4) None of these.



Q81 Pick out the **incorrect** statement.

- (1) The standard free energy of formation of elements is zero.
- (2) A process that leads to increase in free energy will be spontaneous.
- (3) $\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surrounding}}$
- (4) Enthalpy of combustion is always negative.

Q82 Hybridisation and shape of XeF_2 molecule is:

- (1) sp^3d , Angular
- (2) sp^3d , See-Saw
- (3) sp^3d , linear
- (4) sp^3d , Tetrahedral

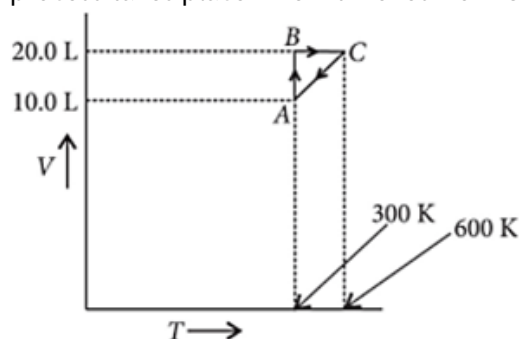
Q83 The activation energy of a reaction can be determined from the slope of which of the following graphs?

- (1) $\frac{T}{\ln K}$ vs $\frac{1}{T}$
- (2) $\ln K$ vs T
- (3) $\frac{\ln K}{T}$ vs T
- (4) $\ln K$ vs $\frac{1}{T}$

Q84 The heats of combustion of CH_4 and C_4H_{10} are -890.3 kJ/mol and -2878.7 kJ/mol respectively. The fuel which is more efficient is:

- (1) methane
- (2) butane
- (3) both have equal efficiency
- (4) none of these

Q85 This graph expresses the various steps of the system containing 1 mole of gas. Which type of process takes place when it moves from C to A?

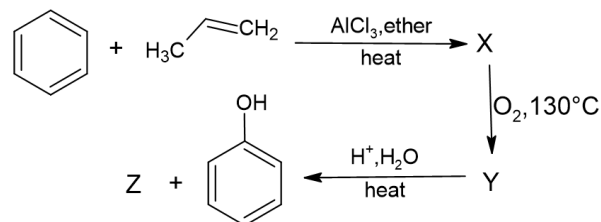


- (1) Isochoric
- (2) Isobaric
- (3) Isothermal
- (4) Cyclic

Q86

SECTION - B (CHEMISTRY)

The products X and Z in the following sequence are:



- (1) isopropylbenzene and acetone
- (2) cumeneperoxide and acetone
- (3) isopropylbenzene and isopropylalcohol
- (4) phenol and acetone

Q87 A compound (A) when treated with CH_3MgI in the presence of dry ether followed by hydrolysis gives alcohol (B) of formula $\text{C}_5\text{H}_{11}\text{OH}$, which on oxidation gives pentan-2-one. What is (A)?

- (1) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$
- (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
- (3) $\text{CH}_3\text{CH}_2\text{COCH}_3$
- (4) $\text{CH}_3\text{CH}_2\text{CHO}$

Q88 Match the reaction in **List-I** with relation between ΔH and ΔU in **List-II**.

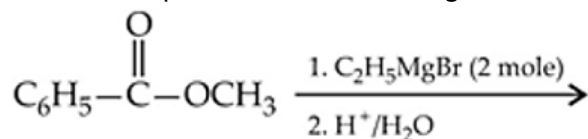
	List-I		List-II
(A)	$\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$	(I)	$\Delta H = \Delta U + RT$
(B)	$\text{N}_2\text{(g)} + 3\text{H}_2\text{(g)} \rightarrow 2\text{NH}_3\text{(g)}$	(II)	$\Delta H = \Delta U$
(C)	$\text{NH}_4\text{HS(s)} \rightarrow \text{NH}_3\text{(g)} + \text{H}_2\text{S(g)}$	(III)	$\Delta H = \Delta U - 2RT$
(D)	$\text{PCl}_5\text{(g)} \rightarrow \text{PCl}_3\text{(g)} + \text{Cl}_2\text{(g)}$	(IV)	$\Delta H = \Delta U + 2RT$

- (1) A-IV, B-II, C-III, D-I



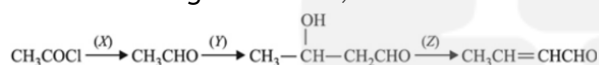
- (2) A-III, B-II, C-I, D-IV
 (3) A-IV, B-I, C-II, D-III
 (4) A-II, B-III, C-IV, D-I

Q89 Which is the product of the following reactions?



- (1) $\text{C}_6\text{H}_5-\text{CH}_2-\text{OH}$
 (2) $\text{C}_6\text{H}_5-\underset{\text{OH}}{\text{CH}}-\text{C}_2\text{H}_5$
 (3) $\text{C}_6\text{H}_5-\underset{\text{OH}}{\overset{\text{C}_2\text{H}_5}{\text{C}}}-\text{C}_2\text{H}_5$
 (4) $\text{C}_6\text{H}_5-\overset{\text{O}}{\parallel}{\text{C}}-\text{OC}_2\text{H}_5$

Q90 In the following conversion,



reagent X,Y and Z are:

- (1) (X) $\text{H}_2, \text{Pd}/\text{BaSO}_4$, (Y) dil. NaOH , (Z) heat
 (2) (X) NaOH , (Y) Hydrolysis, (Z) heat
 (3) (X) I_2/NaOH , (Y) LiAlH_4 , (Z) H_3O^+
 (4) (X) CrO_3 , (Y) Warm, (Z) CO_2
- Q91** Stephen's reduction converts:
- (1) acyl chloride to aldehyde
 (2) cyanide to aldehyde
 (3) cyanide to carboxylic acid
 (4) cyanide to ketones

Q92 The rate constant for the first order decomposition of ethylene oxide into CH_4 and CO is described by:

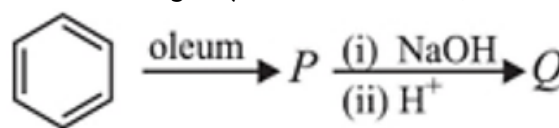
$$\log k = 14.34 - \frac{12.5 \times 10^4 \text{ K}}{T}$$

Then the activation energy of the reaction is:

- (1) $2.39 \times 10^5 \text{ kJ mol}^{-1}$

- (2) $2.39 \times 10^3 \text{ kJ mol}^{-1}$
 (3) $4.78 \times 10^5 \text{ kJ mol}^{-1}$
 (4) $4.78 \times 10^2 \text{ kJ mol}^{-1}$

Q93 In the following sequence of reactions,



the compound Q formed will be:

- (1) aniline
 (2) phenol
 (3) benzaldehyde
 (4) benzene sulphonic acid.

Q94 Activation energy (E_a) and rate constants (k_1 and k_2) of a chemical reaction at two different temperature (T_1 and T_2) are related by:

- (1) $\ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$
 (2) $\ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$
 (3) $\ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} + \frac{1}{T_1} \right)$
 (4) $\ln \frac{k_2}{k_1} = +\frac{E_a}{R} \left(\frac{1}{T_1} + \frac{1}{T_2} \right)$

Q95 What is the heat of hydration of CuSO_4 if heat of solution of anhydrous CuSO_4 in a large volume of water is -15.90 kcal . and that of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is 2.75 kcal . at 18°C

- (1) -18.65 kcal .
 (2) -13.15 kcal .
 (3) $+13.15 \text{ kcal}$.
 (4) -23.15 kcal .

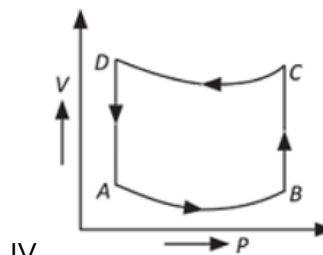
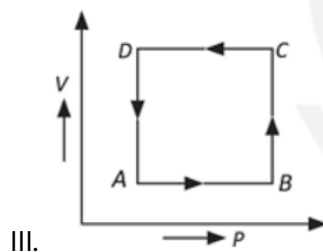
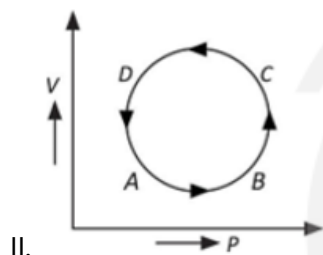
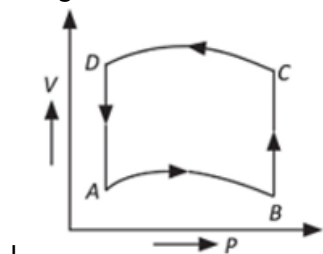
Q96 For first order reaction, how is time required for 99% completion related to the time required for the completion of 90% of reaction?

- (1) $t_{90\%} = 2 \times t_{99\%}$
 (2) $t_{90\%} = 4 \times t_{99\%}$
 (3) $t_{99\%} = \frac{1}{2} \times t_{90\%}$
 (4) $t_{99\%} = 2 \times t_{90\%}$



- Q97** Enthalpy of neutralisation of HCl with NaOH is x . The heat evolved when 500 mL of 2 N HCl are mixed with 250 mL of 4 N NaOH will be:
- (1) $500x$ (2) $100x$
(3) x (4) $10x$

- Q98** In diagrams (I to IV), variation of volume with changing pressure is shown. A gas is taken along the path ABCD. The change in internal energy of the gas will be:



- (1) positive in all the cases (I) to (IV)
(2) positive in cases (I), (II), (III) but zero in case (IV)
(3) negative in cases (I), (II), (III) but zero in case (IV)

- (4) zero in all the cases

- Q99** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R,

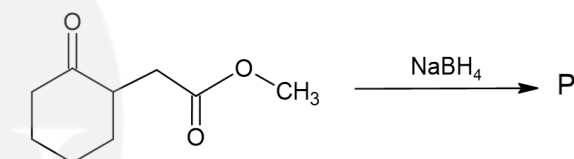
Assertion A: KMnO_4 is an oxidising agent in neutral, acidic and basic medium.

Reason R: Equivalent mass of KMnO_4 in acidic medium is 31.6

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

- Q100** Identify the product P in the following reaction:



- (1)
- (2)
- (3)
- (4)



Which one of the following statement is **correct**?

- (1) Geitonogamy involves the pollen and stigma of flowers of different plants.
- (2) Cleistogamous flowers are always autogamous.
- (3) Xenogamy occurs only by wind pollination.
- (4) Chasmogamous flowers do not open at all.

Q102 Identify from the following which of the products are formed by bacteria.

- A. Penicillin
- B. Butyric acid
- C. Fermentation of dosa dough
- D. Fermentation of idli dough
- E. Rum

Choose the most appropriate answer from the options given below:

- (1) Only A, B, C and E are correct
- (2) Only A, B and D are incorrect
- (3) Only A, B and C are correct
- (4) Only B, C and D are correct

Q103 The genotypes of a husband and wife are $I^A I^B$ and $I^A i$. Among the blood types of their children, how many different genotypes and phenotypes are possible?

- (1) 3 genotypes; 4 phenotypes
- (2) 4 genotypes; 3 phenotypes
- (3) 4 genotypes; 4 phenotypes
- (4) 3 genotypes; 3 phenotypes

Q104 Read the following statements and mark the **correct** option.

- A. *Viola* can show autogamy.
- B. Common passy can undergo cross pollination.
- C. *Oxalis* has chasmogamous flower.
- D. *Commelina* has both types of flower cleistogamous and chasmogamous.

Choose the most appropriate answer from the options given below:

- (1) A and C only

- (2) B and D only
- (3) A and D only
- (4) A, B, C and D

Q105 Flocs in sewage treatment plant are:

- (1) primary sludge produced.
- (2) anaerobic bacteria.
- (3) a mesh like structure formed by association of bacteria and fungal filaments.
- (4) the effluent in primary treatment.

Q106 Which of the following microbes is **correctly** paired with its function?

- (1) *Aspergillus niger* - Production of lactic acid
- (2) *Trichoderma polysporum* - production of lipase
- (3) *Saccharomyces cerevisiae* - Production of citric acid
- (4) Methanogenic bacteria - Biogas formation

Q107 Select the **correct** statements for benefits of mycorrhizae.

- A. Resistance against root-borne pathogens.
- B. Tolerance to salinity and drought.
- C. Helps in absorption of phosphorous.
- D. Overall increase in the plant growth and development.

Choose the most appropriate answer from the options given below:

- (1) A and C only
- (2) B and D only
- (3) A and D only
- (4) A, B, C and D

Q108 Starting from the innermost wall layer the **correct** sequence of wall layer in an anther are:

- (1) Epidermis → Tapetum → Endothecium → Middle layers.
- (2) Endothecium → Middle layers → Epidermis → Tapetum



- (3) Epidermis → Endothecium → Middle layers
→ Tapetum
- (4) Tapetum → Middle layers → Endothecium →
Epidermis

Q109 Select the **incorrect** option from the following statements.

- (1) Down's syndrome is chromosomal disorder.
- (2) Sickle cell anaemia is an autosomal recessive trait.
- (3) Sickle cell anaemia is quantitative problem of synthesising incorrectly functioning globin.
- (4) Cystic fibrosis is a Mendelian disorder.

Q110 Match **List-I** with **List-II**.

List-I		List-II	
(A)	Methanogens	(I)	Microbial biocontrol agent
(B)	Organic wastes	(II)	Penicillin
(C)	<i>Bacillus thuringiensis</i>	(III)	Biochemical oxygen demand
(D)	<i>Penicillium notatum</i>	(IV)	<i>Methanobacterium</i>

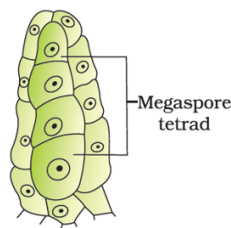
Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-IV, B-III, C-I, D-II
- (3) A-III, B-IV, C-II, D-I
- (4) A-II, B-I, C-IV, D-III

Q111 KVIC stands for:

- (1) Khadi and Village Institution Commission
- (2) Khadi and Vellore Institution Commission
- (3) Khadi and Village Industries Commission
- (4) Kanpur and Vellore Industries Commission

Q112 In angiosperms the given process of megasporogenesis occurs at:



- (1) central cell.
- (2) antipodals.
- (3) micropylar end.
- (4) chalazal end.

Q113 Given below are two statements.

Statement I: The number of ovule in orchids is one.

Statement II: *Papaver* has apocarpous pistil.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct and Statement-II is incorrect.
- (2) Statement-I is incorrect and Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

Q114 A man has enlarged breasts on the body and sex chromosomal formula XXY. He then suffers from:

- (1) Down's syndrome.
- (2) sickle cell anaemia.
- (3) Turner's syndrome.
- (4) Klinefelter's syndrome.

Q115 Chromosomal theory of inheritance:

- (1) was proposed independently by Sutton and Boveri.
- (2) shows parallelism between behaviour of chromosomes and behaviour of genes.
- (3) was experimentally verified by T.H. Morgan.
- (4) All of these



Q116 Given below are two statements:

Statement I: Pollination is done as only egg is non motile.

Statement II: Depending on the source of pollen, pollination can be divided into two types.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

Q117 Which of the following is a **mismatched** pair?

- (1) Microsporangium – Pollen sac
- (2) Megasporangium – Ovule
- (3) Intine-Cellulose and pectin
- (4) Rosaceae - Viability of pollen for 30 minutes

Q118 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Organic farmers work to create a system where pests are kept at manageable level not completely eradicated.

Reason R: Key belief of organic farmers is that biodiversity furthers health.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q119 Which of the following statements is **incorrect** regarding fermentation?

- (1) *Propionibacterium* is used to ferment the cheese.
- (2) The puffed-up appearance of dough is due to the production of CO₂ gas.
- (3) Brewer's yeast is used in production of acetic acid.
- (4) Toddy is made by fermenting sap from palms.

Q120 Given below are two statements.

Statement I: Phenotypic ratio of monohybrid cross in F₂ generation is 3:1.

Statement II: Genotypic ratio of monohybrid cross in F₂ generation is 1:2:1.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct and Statement II is incorrect.
- (2) Statement I is incorrect and Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

Q121 Antibiotics are used to treat diseases like:

- (1) plague and kusht rug only
- (2) whooping cough and gal ghotu only
- (3) leprosy and kali khansi only
- (4) All of these

Q122 Which of the following statements are **correct**?

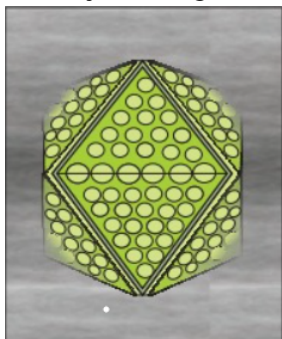
- A. *Nucleopolyhedroviruses* has narrow spectrum insecticidal applications.
- B. BOD is the amount of oxygen that would be consumed if all the inorganic matter in one litre of water were oxidized by bacteria.
- C. *Trichoderma sp.* are free living fungi.
- D. Baculoviruses are pathogens that attack insects and other arthropods.
- E. Bt-cotton is being cultivated in some states of our country.



Choose the most appropriate answer from the options given below:

- (1) Only B and C are correct
- (2) Only A, B and C are incorrect
- (3) Only A, C, D and E are correct
- (4) Only B and D are correct

Q123 Identify the diagram of virus given below.



- (1) Rod shaped virus
- (2) Adenovirus
- (3) Tobacco mosaic virus
- (4) Bacteriophage

Q124 Microsporogenesis occurs:

- (1) inside the anther through mitotic division.
- (2) inside the ovule through meiotic division.
- (3) in sporogenous tissue through meiotic division.
- (4) in thalamus.

Q125 The people who have undergone myocardial infarction are given _____ for the removal of clots from the blood vessels.

- (1) streptokinase (2) lipases
- (3) pectinase (4) statins

Q126 Select the **incorrect** statements.

- A. It has become a fashion in recent years to use pollen tablets as food supplements.
- B. In western countries, a large number of pollen products in the form of tablets and syrups are available in the market.

C. It is possible to store pollen grains of a large number of species for years in liquid oxygen.

D. Pollen may cause pollen allergy.

Choose the most appropriate answer from the options given below:

- (1) A, B and C only
- (2) B, C and D only
- (3) A and D only
- (4) C only

Q127 Match **List-I** with **List-II**:

List-I		List-II	
(A)	Integument	(I)	Stalk of ovule
(B)	Hilum	(II)	Region where body of ovule fuses with funicle
(C)	Funicle	(III)	One or two protective layers of ovule
(D)	Chalaza	(IV)	Basal part of ovule

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-I, B-III, C-II, D-IV
- (3) A-II, B-IV, C-III, D-I
- (4) A-I, B-II, C-III, D-IV

Q128 Identify the benefits of curd.

- A. Source of B₁₂ vitamin.
- B. Increase of disease causing microbes in gut.
- C. Immunosuppressive agent.
- D. Biocontrol agent.

Choose the most appropriate answer from the options given below:

- (1) A, B and C only
- (2) A, B and D only
- (3) A and C only
- (4) A only



Q129 Dioecy prevents:

- (1) geitonogamy but not autogamy.
- (2) both autogamy and geitonogamy.
- (3) autogamy but not geitonogamy.
- (4) both geitonogamy and xenogamy.

Q130 When pink and white flowered plants of snapdragon are crossed then phenotypic ratio in offspring's will be:

- (1) 1 : 2 : 1. (2) 3 : 1.
- (3) 1 : 1 : 1 : 1. (4) 1 : 1.

Q131 Queen Victoria shows a number of haemophilic descendent as she was:

- (1) haemophilic.
- (2) a carrier of the disease.
- (3) having homozygous recessive gene on both the sex chromosome for this disease.
- (4) not a carrier.

Q132 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Blue green algae serves as an important biofertiliser in paddy fields.

Reason R: Blue green algae can fix atmospheric nitrogen as well as adds organic matter to soil.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q133 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Thalassaemia is due to either mutation or deletion.

Reason R: Beta thalassaemia is controlled by two genes on chromosome 11.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q134 During sewage treatment, biogas is produced which include:

- (1) CH₄ and H₂S only
- (2) CH₄ and CO₂ only
- (3) CO₂ and H₂S only
- (4) CH₄, CO₂ and H₂S.

Q135 How many of the following are **correct**.

A. Mendel selected 7 true breeding pea plant varieties.

B. Mendel conducted artificial pollination experiments using several true-breeding pea lines.

C. A true breeding line shows the stable trait inheritance.

D. Mendel applied statistical methods and mathematical logic for analysing the results.

Choose the most appropriate answer from the options given below:

- (1) Two (2) Four
- (3) Three (4) One

Q136

SECTION - B (BOTANY)

Which of the following alcoholic drinks are produced without distillation.

- (1) Wine and whisky
- (2) Whisky and rum
- (3) Beer and brandy
- (4) Wine and beer



Q137 How many of the following show water pollination?

- A. Water lily
- B. *Zostera*
- C. *Vallisneria*
- D. *Hydrilla*

Choose the most appropriate answer from the options given below:

- (1) One
- (2) Two
- (3) Three
- (4) Four

Q138 Select **correct** statement for colour blindness.

- (1) Colour blindness is an allele carried on one of the mother's autosomal chromosomes.
- (2) Colour blindness is an allele carried on the Y-chromosome because more males have this genetic disorder than females.
- (3) Colour blindness is an allele carried on the X-chromosome and can be directly inherited by the son from the father or the mother.
- (4) Colour blindness is carried on the X-chromosome and can only be inherited by the son if the mother is a carrier.

Q139 Observe the sex determination in the following organism.

- A. Human males = XY
- B. Female hen = ZW
- C. Male birds = ZZ
- D. Male grasshopper = XO

Which of the following combination is **correct**:

- (1) A and C only
- (2) C and D only
- (3) A and D only
- (4) A, B, C and D

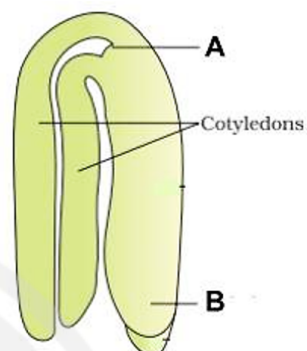
Q140 Segregation of Mendelian factors occurs during:

- (1) anaphase I.
- (2) anaphase II.
- (3) metaphase II.
- (4) metaphase I.

Q141 A cross between pure tall pea plant with green pods and dwarf pea plant with yellow pods will produce how many dwarf plants in F_2 out of 16 ?

- (1) Nine
- (2) Three
- (3) Four
- (4) One

Q142 Identify the part **A** or **B** labelled in the following diagram of typical dicot embryo and select the **correct** option.



- (1) A-Coleorhizae
- (2) A-Coleoptile
- (3) B-Radicle
- (4) B-Plumule

Q143 Match **List-I** with **List-II**.

List-I		List-II	
(A)	Alfred Sturtevant	(I)	Mapped position of genes
(B)	Henking	(II)	X-body
(C)	Correns	(III)	Involved in rediscovery of Mendel's law on the inheritance of characters
(D)	Morgan	(IV)	Linkage

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-I, B-III, C-II, D-IV
- (3) A-II, B-IV, C-III, D-I
- (4) A-I, B-II, C-III, D-IV



Q144 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Ganga action plan and Yamuna action plan is proposed to save rivers from water pollution.

Reason R: According to Ganga and Yamuna action plan, treated sewage cannot be discharged into these rivers.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q145 A normal woman whose father was colour blind, marries a normal man. What kinds of children can be expected and in what proportion?

- (1) All daughters normal, 50% of sons colourblind
- (2) All daughters normal, all sons colourblind
- (3) 50% daughters colourblind, all sons normal
- (4) All daughters colourblind, all sons normal

Q146 The product formed due to double fertilisation respectively are:

- (1) embryo and perisperm.
- (2) zygote and scutellum.
- (3) zygote and primary endosperm nucleus.
- (4) endosperm and cotyledons.

Q147 Given below are two statements:

Statement I: Morgan carried out several monohybrid cross to study genes that were showing independent assortment.

Statement II: Genetic maps can be used in human genome project.

In the light of the above statements, choose the most appropriate answer from the options given

below:

- (1) Statement-I is correct and Statement-II is incorrect.
- (2) Statement-I is incorrect and Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

Q148 Select the **correct** statements related to endosperm.

- A. White kernel in coconut is free-nuclear endosperm.
- B. The coconut water is the cellular endosperm.
- C. The central cell after triple fusion becomes the primary endosperm cell.
- D. In castor seeds endosperm is present.

Choose the most appropriate answer from the options given below:

- (1) A and C only
- (2) C and D only
- (3) A and D only
- (4) B and C only

Q149 Select the **incorrect** statements for fruit.

- A. Strawberry is a false fruit.
- B. Guava is a fleshy fruit.
- C. Ovary develops into a fruit.
- D. Cashew is a true fruit.

Choose the most appropriate answer from the options given below:

- (1) A and B only
- (2) A, B and D only
- (3) A, B and C only
- (4) D only

Q150 Choose the **correct** sequence for performing artificial hybridisation.

- A. Rebagging
- B. Bagging
- C. Selection of parents



- D. Emasculation
E. Dusting the pollen on stigma
- (1) $C \rightarrow D \rightarrow B \rightarrow E \rightarrow A$
(2) $D \rightarrow C \rightarrow E \rightarrow B \rightarrow A$
(3) $E \rightarrow B \rightarrow A \rightarrow F \rightarrow C$
(4) $C \rightarrow D \rightarrow E \rightarrow B \rightarrow A$

Q151

SECTION - A (ZOOLOGY)

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

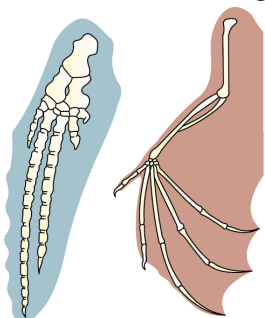
Assertion A: Hepatitis-B is a sexually transmitted disease.

Reason R: Hepatitis- B can be spread through protected sexual contact.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

Q152 Refer to the following diagram.



X

Y

Which of the following represents X and Y **correctly**?

- (1) Convergent evolution
(2) Industrial melanism
(3) Homologous organs
(4) Adaptive radiation

Q153 Choose the **incorrect** statement w.r.t human sperm.

- (1) The human male ejaculates about 200 to 300 million sperms during a coitus.
(2) The plasma membrane envelops the whole body of sperm.
(3) A haploid nucleus is present in the neck of sperms.
(4) The tail of sperm helps in motility.

Q154 How many million years ago did *Dryopithecus* and *Ramapithecus* exist?

- (1) 5 (2) 7
(3) 20 (4) 15

Q155 Given below are two statements:

Statement I: In disruptive natural selection, more individuals acquire peripheral character value at both ends of the distribution curve.

Statement II: Cave paintings by Pre-historic humans can be seen at Bhimbetka rock shelter.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q156 Given below are two statements:

Statement I: Vas deferens receives a duct from seminal vesicle and opens into urethra as the ejaculatory duct.

Statement II: Parts of the female reproductive system alongwith a mammary gland are integrated structurally and functionally to support the processes of ovulation, fertilisation, pregnancy, birth and child care.



In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

Q157 Match the **List-I** with **List-II**.

List-I		List-II	
(A)	Hormone releasing IUDs	(I)	Progestasert
(B)	Copper releasing IUDs	(II)	Multiload 375
(C)	Non-medicated IUDs	(III)	Lippes loop
(D)	Barrier contraceptive method	(IV)	Cervical caps

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-III, B-I, C-II, D-IV
- (3) A-I, B-II, C-III, D-IV
- (4) A-III, B-II, C-I, D-IV

Q158 Which of the organisms have evolved due to changes in environment brought about by anthropogenic action?

- I. Darwin's Finches of Galapagos islands.
- II. Herbicide resistant weeds.
- III. Drug resistant bacteria.
- IV. Australian marsupials.

- (1) I and III only
- (2) II and III only
- (3) IV only
- (4) I only

Q159 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Evolution is a stochastic process.

Reason R: Evolution is based on chance events in nature and chance mutation in the organisms.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false
- (2) A is false but R is true.
- (3) Both A and R are true but R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q160 Seminal plasma is contributed by:

- I. seminal vesicle.
- II. prostate gland.
- III. urethra.
- IV. bulbourethral gland.

Choose the *most appropriate* answer from the options given below:

- (1) I and II only
- (2) I, II and IV only
- (3) II, III and IV only
- (4) I and IV only

Q161 Read the following statements (I-V) about MTP.

- I. It is the intentional or voluntary termination of pregnancy before full term.
- II. It is used to get rid of unwanted pregnancies due to casual unprotected intercourse.
- III. Majority of the MTPs are performed illegally by unqualified quacks.
- IV. MTPs are safe during the third trimester.
- V. Government of India legalised MTP in 1992.

Choose the *most appropriate* answer from the options given below:

- (1) Only I and II are correct.
- (2) Only II and V are correct.
- (3) Only I, II and IV are correct.



(4) Only I, II and III are correct.

Q162 In human beings, at the end of the first meiotic division, the primary spermatocytes differentiate into the:

- (1) spermatids.
- (2) spermatogonia.
- (3) spermatozoa.
- (4) secondary spermatocytes.

Q163 Given below are two statements:

Statement I: Darwinian variations are small and directional.

Statement II: According to Hardy Weinberg equilibrium, allele frequencies in a population are unstable and is constant from generation to generation.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

Q164 Which of the following options best represents chemical evolution?

- (1) Formation of inorganic molecules from organic constituents.
- (2) Formation of organic molecules from inorganic constituents.
- (3) Formation of inorganic molecules from inorganic constituents.
- (4) Formation of organic molecules from organic constituents.

Q165 If the number of chromosomes in a primary spermatocyte is 16. What will be the total

number of chromosomes in each of the secondary spermatocyte?

- (1) 16
- (2) 24
- (3) 32
- (4) 8

Q166 Which of the following options is **correct** w.r.t the highest brain capacity from the evolution of human beings?

- (1) *Homo habilis*
- (2) *Homo erectus*
- (3) Neanderthal man
- (4) Both (1) and (2)

Q167 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Branching descent and natural selection are the two key concepts of Darwinian Theory of Evolution.

Reason R: Fitness is the end result of the ability to adapt and get selected by nature according to Darwin.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true but R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q168 Identify the fossil of man who showed the following characteristics.

- A. Brain capacity of 1400 cc.
- B. Used hides to protect their body.
- C. Buried their dead bodies.

In the light of above statements, choose the **correct** answer from the options given below:

- (1) *Homo erectus*
- (2) Neanderthal man
- (3) *Homo habilis*
- (4) *Australopithecus*



Q169 Which of the following is the **correct** sequence of transport of milk in a mammary gland?

- (1) Mammary tubule → Mammary duct → Mammary ampulla → Lactiferous duct
- (2) Mammary duct → Mammary tubule → Lactiferous duct → Mammary ampulla
- (3) Mammary duct → Mammary tubule → Mammary ampulla → Lactiferous duct
- (4) Mammary ampulla → Mammary tubule → Mammary duct → Lactiferous duct

Q170 'Panspermia' is still a favourite idea for some astronomers in context of origin of life. According to this concept:

- (1) life was created by God.
- (2) life originated from non-living material like mud and straw.
- (3) units of life called spores were transferred to different planets including earth.
- (4) life arose from some pre-existing life.

Q171 Read the following statements (A-E).

- A. The presence of gill slits in the embryos of all vertebrates support the theory proposed by Ernst Haeckel.
- B. Around 65 million years ago, some of the land reptiles went back to water and evolved into fish like reptiles, called *Ichthyosaurs*.
- C. Dinosaurs became extinct 200 million years ago.
- D. The first mammals were like shrews.
- E. Due to the continental drift, pouched mammals of Australia survived because of lack of competition from any other mammal.

Which of the following statements are **incorrect**?

- (1) A and B only
- (2) B and C only
- (3) C, D and E only
- (4) A, C, D and E only

Q172 Which of the following conditions were similar as that of primitive earth conditions?

- (1) Low temperature, volcanic storms, atmosphere rich in oxygen.
- (2) Low temperature, volcanic storms, reducing atmosphere
- (3) High temperature, volcanic storms, non reducing atmosphere.
- (4) High temperature, volcanic storms, reducing atmosphere.

Q173 Which of the following statement is **incorrect** regarding implants?

- (1) It contains progestogens or combinations of estrogen and progestogen.
- (2) It is effective for longer period as compared to pills.
- (3) They are used by both males and females.
- (4) All of these

Q174 Read the following statements (I-V).

- I. Statutory ban on amniocentesis for sex-determination to legally check increasing menace of female foeticides.
- II. India was amongst the first countries in the world to initiate action plans and programmes at a national level to attain total reproductive health as a social goal.
- III. Spermicidal creams, jellies and foams are usually used along with barriers to decrease their contraceptive efficiency.
- IV. Genital herpes is completely curable if detected early and treated properly.
- V Progestasert is a type of hormone releasing IUD.

Choose the *most appropriate* answer from the options given below:

- (1) Only I and II are incorrect.
- (2) Only II and III are incorrect.
- (3) Only III and IV are incorrect.
- (4) Only I and V are incorrect.



Q175 Complete the given analogy and choose the **correct** option.

Bobcat : Tasmanian tiger cat :: _____ : Spotted cuscus.

- (1) Numbat (2) Wolf
(3) Lemur (4) Anteater

Q176 Read the following statements (I-IV):

I. The white-winged moths were completely wiped out after industrialisation.

II. Since lichens did not grow in polluted area, the number of melanised moths got reduced.

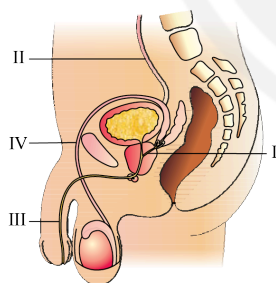
III. Karl Ernst von Baer noted that embryos never pass through the adult stages of other animals.

IV. Cheetah and all mammals share similarities in the pattern of bones of forelimbs.

Which of the following statements is **correct**?

- (1) I and II only
(2) I, II and III
(3) III and IV only
(4) II and III only

Q177 Given below is a diagrammatic sectional view of male reproductive system. Identify the duct that ascends to the abdomen and loops over the urinary bladder.



- (1) I (2) II
(3) III (4) IV

Q178 Given below are two statements:

Statement I: The testes are situated outside the abdominal cavity within a pouch called scrotum.

Statement II: The chorionic villi and uterine

tissue become interdigitated with each other and jointly form umbilical cord.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q179 Complete the given analogy and choose the **correct** option.

Progesterone: Maintenance of endometrium:: _____: Parturition

- (1) FSH (2) LH
(3) Oxytocin (4) Estrogen

Q180 Given below are two statements:

Statement I: Zygote intrafallopian transfer is an *in-vitro* fertilisation technique.

Statement II: The menstrual flow results due to breakdown of endometrial lining of uterus.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q181 How many of the following hormones given in box is/are **not** produced by placenta?

hCG, Progesterone, FSH, LH, Estrogen, hPL, Thyroxine



- (1) One (2) Two
(3) Three (4) Four

Q182 Given below are two statements:

Statement I: By the time of 500 mya, invertebrates were formed and active.

Statement II: Earth is about 4000 years old according to special creation theory.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q183 Match List-I with List-II.

List - I		List - II	
(A)	Mole	(I)	Fossils discovered in Java.
(B)	<i>Homo erectus</i>	(II)	Lived in between 10,000-40,000 years back.
(C)	Neanderthal man	(III)	Marsupial mole
(D)	Anteater	(IV)	Numbat

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-II, B-III, C-I, D-IV
(4) A-III, B-I, C-II, D-IV

Q184 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Cu-T is a copper releasing IUD.

Reason R: Cu^{++} ions released by copper releasing IUDs increase sperm motility and phagocytosis of

the sperms in the uterus.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

Q185 If frequency of a dominant allele in a population which is in genetic equilibrium is 0.7, then calculate frequency of heterozygotes in that population.

- (1) 0.21
(2) 0.24
(3) 0.48
(4) 0.42

Q186

SECTION - B (ZOOLOGY)

____A____ and ____B____ forms the birth canal.

Choose the options which fill in the blanks **correctly**.

- (1) A - cervix, B - uterus
(2) A - fallopian tube, B - uterus
(3) A - cervix, B - vagina
(4) A - uterus, B - fallopian tube

Q187 Match List-I with List-II.

List-I		List-II	
(A)	Condom	(I)	Non-steroidal pill
(B)	Saheli	(II)	Hormone releasing IUD
(C)	LNG-20	(III)	Barrier contraceptive method
(D)	Withdrawal method	(IV)	Natural contraceptive method



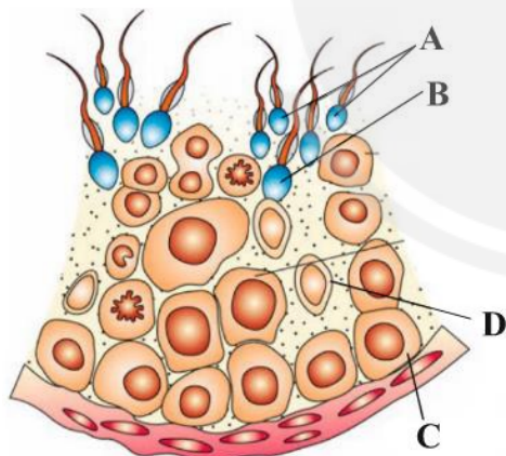
Choose the **correct** answer from the options given below:

- (1) A-I, B-IV, C-II, D-III
- (2) A-II, B-III, C-I, D-IV
- (3) A-III, B-II, C-IV, D-I
- (4) A-III, B-I, C-II, D-IV

Q188 Which of the following statements are **incorrect**?

- (1) Increased production of hormones during pregnancy is essential for supporting the foetal growth, metabolic changes in the mother and maintenance of pregnancy.
- (2) Pills inhibit release of secondary oocytes and attachments of blastocyst to the endometrium layer of uterus.
- (3) The second meiotic division of primary oocyte is unequal and results in the formation of a second polar body and a haploid ootid.
- (4) Gonorrhoea is a venereal disease that are not transmitted through sexual intercourse.

Q189 In the given below diagram of seminiferous tubules, some parts are labelled as **A, B, C** and **D**.



Which of the above labelled cells undergo spermiogenesis?

- (1) A
- (2) B
- (3) D
- (4) C

Q190 Given below are two statements:

Statement I: Inability to conceive or produce

children even after 2 years of unprotected sexual cohabitation is called infertility.

Statement II: 'Nirodh' is a popular brand of condom for the female.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement I is correct, but Statement II is incorrect.
- (2) Statement I is incorrect, but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

Q191 Read the following statements w.r.t cervical caps.

- I. It is a barrier method of contraception.
- II. It covers the vagina only during coitus.
- III. It protects the user from contracting STDs.
- IV. It is reusable.

Which of the above statements are **correct**?

- (1) I and IV only
- (2) I, II and III only
- (3) I, II and IV only
- (4) I, II, III and IV

Q192 Read the following statements (I-IV).

- I. Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region.
- II. The placenta facilitate the supply of oxygen and nutrients to the embryo.
- III. The milk produced during the initial few days of lactation is called colostrum.
- IV. In the later phase of pregnancy, a hormone called relaxin is secreted by the fallopian tube.

Which of the above statements are **correct**?

- (1) I, II and III only
- (2) I and II only
- (3) II, III and IV only
- (4) I, II, III and IV



Q193 Arrange the following plant forms in a sequential order of their evolution and choose the **correct** option.

- I. Psilophyton
- II. Tracheophyte ancestors
- III. Chlorophyte ancestors
- IV. Rhynia-type plants

- (1) III → II → IV → I
- (2) IV → II → III → I
- (3) I → II → III → IV
- (4) II → III → IV → I

Q194 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Oogenesis is different than spermatogenesis.

Reason R: Oogenesis is initiated during the embryonic development stage while spermatogenesis starts at puberty.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q195 Match List-I with List-II.

List-I		List-II	
(A)	Fossils	(I)	Analogous organs
(B)	Theory of abiogenesis	(II)	First human-like being
(C)	Convergent evolution	(III)	Palaeontological evidence

(D)	<i>Homo habilis</i>	(IV)	Living organisms originated from non-living matter
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Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III, D-IV
- (2) A-III, B-IV, C-I, D-II
- (3) A-III, B-IV, C-II, D-I
- (4) A-I, B-IV, C-III, D-II

Q196 All of the following statements are correct, **except**:

- (1) Vasa efferentia is a male accessory duct.
- (2) Spermatogonia is diploid, multiply by mitotic division and increase in number.
- (3) Primary oocyte within the tertiary follicle grows in size and completes its second meiotic division.
- (4) LH and FSH stimulates follicular development as well as secretion of estrogens by the growing follicles.

Q197 Read the following statements (I-V).

- I IUDs are one of most widely accepted methods of contraception in India.
 - II. Statutory raising of marriageable age of the female to 21 years and that of males to 18 years.
 - III. The hormone releasing IUDs make the uterus unsuitable for implantation.
 - IV. IUDs are inserted by doctors or expert nurses in the uterus through vagina.
 - V. Jaundice can be diagnosed by amniocentesis.
- How many of the above statements is/are **incorrect**?

- (1) Two
- (2) One
- (3) Four
- (4) Three

Q198 Which of the following remains blocked after tubectomy?

- (1) Cervix
- (2) Uterine
- (3) Fallopian tube
- (4) Mons pubis



Q199 Which of the following sources (A-E) can lead to evolution?

- A. Mutations
- B. Genetic recombination during gametogenesis
- C. Gene flow
- D. Genetic drift
- E. Natural selection

- (1) A, B and C only
- (2) B, C and E only
- (3) A, B, C and D only
- (4) A, B, C, D and E

Q200 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Adaptive radiation is the evolutionary process in which many species originate from one species in an area and radiate to different places.

Reason R: Darwin's finches represent one of the best examples of adaptive radiation.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.



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